

2-Kernels of Graphs and Digraphs: Six Self-Contained Theorems

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Abstract

A *2-kernel* of a digraph is a set of vertices that is independent and such that every vertex outside it has at least two out-neighbours inside it. This article gives complete proofs of six theorems about 2-kernels: a sufficient condition for two disjoint 2-kernels in a strongly connected digraph, decidability and uniqueness on chordal graphs, the same on digraphs with a chordal underlying graph, a tight lower bound of eight vertices for oriented graphs of minimum out-degree at least two, uniqueness and linear-time decidability on acyclic digraphs, and the fact that the original vertex set of any graph is a 2-kernel of its subdivision. Every supporting fact, including the existence of a simplicial vertex in a chordal graph, is proved from first principles; nothing outside this article is used.

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1 Introduction

Let $D = (V, A)$ be a finite digraph. A set $S \subseteq V$ is a *2-kernel* of D if no arc of D has both ends in S , and every vertex outside S has arcs to at least two distinct vertices of S . An undirected

graph is identified with the digraph obtained by replacing each edge $\{u, v\}$ by the two arcs (u, v) and (v, u) , so that the same definition applies to graphs.

This article proves the following six statements. Each is stated precisely, and with all hypotheses, in the section indicated.

- (i) If a digraph D is strongly connected, every directed cycle of D has even length, and every vertex has out-degree at least 2, then $V(D)$ splits into two non-empty sets, each a 2-kernel of D (Theorem 3.2).
- (ii) On a chordal graph a certain propagation process always terminates with every vertex decided; consequently a chordal graph has at most one 2-kernel, and 2-kernel existence is decidable in polynomial time (Theorem 6.1).
- (iii) The same conclusion holds for every digraph whose underlying graph is chordal, with no assumption on out-degrees (Theorem 7.4).
- (iv) If an oriented graph has minimum out-degree at least 2 and possesses a 2-kernel, then it has at least 8 vertices, and 8 is attained (Theorem 8.1 and Example 8.2).
- (v) An acyclic digraph has at most one 2-kernel, and whether it has one can be decided, and the 2-kernel produced, in linear time (Theorem 9.4 and Corollary 9.5).
- (vi) For every graph G , the vertex set $V(G)$ is a 2-kernel of the subdivision $S(G)$ (Theorem 10.2).

These statements were first observed during a computational study of 2-kernels; the purpose of the present document is to give them proofs that stand on their own. Every lemma used is proved here, and no external theorem is quoted.

2 Preliminaries

All graphs and digraphs in this article are finite. Graphs are simple (no loops, no repeated edges). Digraphs are loopless and have no repeated arcs; between an ordered pair of distinct vertices u, v the arc (u, v) is either present or absent.

2.1 Graphs, digraphs, and the underlying graph

A *graph* $G = (V, E)$ has a vertex set $V = V(G)$ and an edge set $E = E(G)$ of 2-element subsets of V . Vertices u, v are *adjacent* if $\{u, v\} \in E$; we then also write $uv \in E$. The *neighbourhood* of v is $N_G(v) = \{u : uv \in E\}$ and the *degree* of v is $|N_G(v)|$.

A *digraph* $D = (V, A)$ has a vertex set $V = V(D)$ and an *arc* set $A = A(D)$ of ordered pairs of distinct vertices. The *out-neighbourhood* and *in-neighbourhood* of v are

$$N_D^+(v) = \{w : (v, w) \in A\}, \quad N_D^-(v) = \{w : (w, v) \in A\},$$

with *out-degree* $d_D^+(v) = |N_D^+(v)|$ and *in-degree* $d_D^-(v) = |N_D^-(v)|$. The *minimum out-degree* is $\delta^+(D) = \min_{v \in V(D)} d_D^+(v)$; whenever we write $\delta^+(D) \geq 2$ we implicitly assume $V(D) \neq \emptyset$.

The *underlying graph* of a digraph D is the graph $U(D)$ with vertex set $V(D)$ in which u and v are adjacent exactly when $(u, v) \in A(D)$ or $(v, u) \in A(D)$. We write $N_D(v) = N_D^+(v) \cup N_D^-(v)$ for the neighbourhood of v in $U(D)$, and call it the *underlying neighbourhood* of v . Note the containment

$$N_D^+(v) \subseteq N_D(v). \tag{2.1}$$

A digraph D is an *oriented graph* if there are no two vertices u, v with both $(u, v) \in A(D)$ and $(v, u) \in A(D)$. A digraph D is *symmetric* if $(u, v) \in A(D)$ implies $(v, u) \in A(D)$. A graph G is identified with the symmetric digraph on $V(G)$ whose arcs are the pairs (u, v) and (v, u) for each edge $uv \in E(G)$; then $U(D) = G$ and $N_D^+(v) = N_D^-(v) = N_G(v)$ for every v .

2.2 Walks, paths, cycles, connectivity

A *walk* of length ℓ from x to y in a graph is a sequence $x = z_0, z_1, \dots, z_\ell = y$ with $z_{t-1}z_t \in E$ for $1 \leq t \leq \ell$. A *path* is a walk whose vertices are pairwise distinct. A graph is *connected* if it has a path between every two vertices; a *component* of a graph is a maximal set of vertices that induces a connected graph.

A *directed walk* of length ℓ from x to y in a digraph is a sequence $x = z_0, z_1, \dots, z_\ell = y$ with $(z_{t-1}, z_t) \in A$ for $1 \leq t \leq \ell$; it is *closed* if $z_0 = z_\ell$. A *directed cycle* of length k is a closed directed walk $z_0, z_1, \dots, z_k = z_0$ with $k \geq 1$ and $z_0, \dots, z_{k-1}$ pairwise distinct; since digraphs are loopless, every directed cycle has length at least 2. A digraph is *acyclic* (a *DAG*) if it has no directed cycle. A digraph is *strongly connected* if for every ordered pair (u, v) of vertices there is a directed walk from u to v .

Lemma 2.1. *If a graph contains a walk of length ℓ from x to y , then it contains a path from x to y whose vertex set is a subset of the walk's vertex set and whose length is at most ℓ .*

Proof. Induction on ℓ . If the walk's vertices are pairwise distinct it is already a path. Otherwise a vertex z occurs at two positions $p < q$; deleting the portion strictly between these two occurrences yields a walk from x to y of length $\ell - (q - p) < \ell$ whose vertex set is a subset of the original. Apply the inductive hypothesis. $\square$

2.3 Cliques, induced subgraphs, chordal graphs

A *clique* of a graph G is a set of pairwise adjacent vertices. A set of size at most 1 is a clique (the condition is vacuous). For $W \subseteq V(G)$, the *induced subgraph* $G[W]$ has vertex set W and edge set $\{uv \in E(G) : u, v \in W\}$. If D is a digraph we write $U(D)[W]$ for the subgraph of $U(D)$ induced by W .

A *chord* of a cycle $z_0, z_1, \dots, z_k = z_0$ is an edge $z_i z_j$ of the graph with z_i, z_j non-consecutive on the cycle, that is $1 \leq |i - j|$ and $|i - j| \neq 1$ and $\{i, j\} \neq \{0, k - 1\}$. A graph is *chordal* if every one of its cycles of length at least 4 has a chord.

A vertex x of a graph G is *simplicial* if $N_G(x)$ is a clique of G .

2.4 Independent sets, 2-domination, 2-kernels

Let D be a digraph. A set $S \subseteq V(D)$ is *independent* if no arc of D has both ends in S ; equivalently, no edge of $U(D)$ has both ends in S . The set S is *2-dominating* if

$$|N_D^+(v) \cap S| \geq 2 \quad \text{for every } v \in V(D) \setminus S.$$

A *2-kernel* of D is a set that is both independent and 2-dominating.

For a graph G , viewed as its symmetric digraph, this reads: $S \subseteq V(G)$ is a 2-kernel of G if no edge of G has both ends in S and $|N_G(v) \cap S| \geq 2$ for every $v \in V(G) \setminus S$.

Lemma 2.2 (Local constraints on a 2-kernel). *Let S be a 2-kernel of a digraph D , and let $v \in V(D)$. Then:*

- (a) *if $N_D^+(v) \cap S \neq \emptyset$, then $v \notin S$;*
- (b) *if $v \notin S$, then $|N_D^+(v) \cap S| \geq 2$;*
- (c) *$|N_D^+(v) \cap S| \neq 1$;*
- (d) *$v \in S$ if and only if $N_D^+(v) \cap S = \emptyset$.*

Proof. (a) If $v \in S$ and $w \in N_D^+(v) \cap S$, then (v, w) is an arc with both ends in S , contradicting independence. Hence $v \notin S$.

(b) This is the definition of 2-domination applied to $v \notin S$.

(c) If $|N_D^+(v) \cap S| = 1$, then $N_D^+(v) \cap S \neq \emptyset$, so $v \notin S$ by (a), so $|N_D^+(v) \cap S| \geq 2$ by (b): a contradiction.

(d) If $v \in S$, then by the contrapositive of (a), $N_D^+(v) \cap S = \emptyset$. Conversely, if $N_D^+(v) \cap S = \emptyset$, then $|N_D^+(v) \cap S| = 0 < 2$, so by the contrapositive of (b), $v \in S$. $\square$

Remark 2.3. Every 2-kernel S of D is a maximal independent set of $U(D)$: a vertex $v \notin S$ has an out-neighbour in S by Lemma 2.2(b), hence a neighbour in S in $U(D)$, so $S \cup \{v\}$ is not independent. This fact is not used below but explains why the sets constructed in the proofs are forced to be as large as they are.

3 Strongly connected digraphs in which every directed cycle is even

Lemma 3.1 (Cycle decomposition of a closed walk). *Let D be a digraph and let $W = w_0, w_1, \dots, w_L$ be a closed directed walk of length $L \geq 0$ (so $w_0 = w_L$). Then there are directed cycles $C_1, \dots, C_r$ of D (with $r \geq 0$, repetitions allowed) such that $|C_1| + \dots + |C_r| = L$.*

Proof. Strong induction on L .

If $L = 0$, take $r = 0$; the empty sum is 0.

Let $L \geq 1$. If $w_0, w_1, \dots, w_{L-1}$ are pairwise distinct, then W itself is a directed cycle of length L : consecutive vertices are joined by arcs and $w_L = w_0$. Take $r = 1$ and $C_1 = W$.

Otherwise there are indices $0 \leq a < b \leq L - 1$ with $w_a = w_b$. Since $w_a = w_b$ and $(w_b, w_{b+1}) \in A(D)$, also $(w_a, w_{b+1}) \in A(D)$, so

$$W' = w_0, w_1, \dots, w_a, w_{b+1}, w_{b+2}, \dots, w_L$$

is a closed directed walk (it starts and ends at $w_0 = w_L$) of length $a + (L - b) = L - (b - a)$, and

$$W'' = w_a, w_{a+1}, \dots, w_b$$

is a closed directed walk (it starts and ends at $w_a = w_b$) of length $b - a$. Because $1 \leq b - a \leq L - 1$, both W' and W'' have length in $\{1, \dots, L - 1\}$. Apply the inductive hypothesis to each and concatenate the two lists of cycles; the lengths sum to $(L - (b - a)) + (b - a) = L$. $\square$

Theorem 3.2. *Let D be a digraph such that*

- (i) D is strongly connected,
- (ii) every directed cycle of D has even length, and
- (iii) $\delta^+(D) \geq 2$.

Then $V(D)$ can be partitioned into two non-empty sets V_0, V_1 , each of which is a 2-kernel of D . In particular D has two disjoint 2-kernels.

Proof. By (iii), $V(D) \neq \emptyset$; fix a vertex $r \in V(D)$.

Step 1. (Parity of walks from r is well defined.) We claim that any two directed walks from r to the same vertex v have lengths of equal parity. Let P and Q be directed walks from r to v . By (i) there is a directed walk R from v to r . Then P followed by R is a closed directed walk based at r of length $|P| + |R|$, and Q followed by R is one of length $|Q| + |R|$. By Lemma 3.1 each of these lengths is a sum of lengths of directed cycles of D , and by (ii) each such length is even; hence $|P| + |R|$ and $|Q| + |R|$ are both even. Subtracting, $|P| - |Q|$ is even, that is $|P| \equiv |Q| \pmod{2}$.

- Step 2. (The parity function.)** By (i), for every vertex v there is a directed walk from r to v . Define $\text{par}(v) \in \{0, 1\}$ to be the parity of the length of any such walk; this is well defined by Step 1. Note $\text{par}(r) = 0$, using the walk of length 0.
- Step 3. (Every arc reverses parity.)** Let $(u, v) \in A(D)$. Take any directed walk P from r to u and append the arc (u, v) to obtain a directed walk from r to v of length $|P| + 1$. By Step 2, $\text{par}(v) \equiv |P| + 1 \equiv \text{par}(u) + 1 \pmod{2}$, so $\text{par}(v) = 1 - \text{par}(u)$.
- Step 4. (The two classes.)** Put $V_0 = \{v : \text{par}(v) = 0\}$ and $V_1 = \{v : \text{par}(v) = 1\}$. These are disjoint, and their union is $V(D)$ because every vertex has a parity (Step 2). Both are non-empty: $r \in V_0$, and by (iii) the vertex r has an out-neighbour w , which lies in V_1 by Step 3.
- Step 5. (Each class is independent.)** Fix $i \in \{0, 1\}$. If some arc had both ends in V_i , then by Step 3 its two ends would have different parities, contradicting that both lie in V_i . So V_i is independent.
- Step 6. (Each class is 2-dominating.)** Fix $i \in \{0, 1\}$ and set $S = V_i$. Let $v \notin S$, so $\text{par}(v) = 1 - i$, that is $v \in V_{1-i}$. By Step 3 every out-neighbour of v has parity i , so $N_D^+(v) \subseteq V_i = S$. Hence

$$|N_D^+(v) \cap S| = |N_D^+(v)| = d_D^+(v) \geq \delta^+(D) \geq 2.$$

So S is 2-dominating.

- Step 7. (Conclusion.)** By Steps 5 and 6, each of V_0 and V_1 is independent and 2-dominating, hence a 2-kernel of D . By Step 4 they are non-empty; they are disjoint and partition $V(D)$.

□

4 The structure of chordal graphs

This section proves, from the definition of chordality alone, that every non-empty chordal graph has a simplicial vertex. Three preparatory lemmas come first.

Lemma 4.1 (Heredity). *If G is a chordal graph and $W \subseteq V(G)$, then $G[W]$ is chordal.*

Proof. Let Z be a cycle of $G[W]$ of length at least 4. Its edges are edges of $G[W]$, hence edges of G , so Z is also a cycle of G . Since G is chordal, Z has a chord xy in G : an edge of G joining two vertices of Z that are non-consecutive on Z . Both $x, y \in W$, so $xy \in E(G[W])$, and xy is a chord of Z in $G[W]$. Thus every cycle of $G[W]$ of length at least 4 has a chord in $G[W]$, so $G[W]$ is chordal. □

Lemma 4.2 (Shortest paths are induced). *Let $P = x_0, x_1, \dots, x_k$ be a path in a graph G such that no path from x_0 to x_k has fewer than k edges. Then $x_i x_j \in E(G)$ implies $|i - j| = 1$.*

Proof. Suppose $x_i x_j \in E(G)$ with $i < j$ and $j - i \geq 2$. Then

$$x_0, x_1, \dots, x_i, x_j, x_{j+1}, \dots, x_k$$

is a walk from x_0 to x_k with $i + 1 + (k - j) = k - (j - i) + 1 \leq k - 1$ edges. By Lemma 2.1 there is a path from x_0 to x_k with at most $k - 1$ edges, contradicting the hypothesis on P . □

Lemma 4.3 (Minimal separators in a chordal graph are cliques). *Let G be a chordal graph and let a, b be distinct non-adjacent vertices of G . Consider the family of all sets $T \subseteq V(G) \setminus \{a, b\}$ such that a and b lie in different components of $G - T$. This family is non-empty; let S be an inclusion-minimal member of it. Then S is a clique of G .*

Proof. The family is non-empty: for $T = V(G) \setminus \{a, b\}$, the graph $G - T$ has vertex set $\{a, b\}$ with no edge (as a, b are non-adjacent), so a and b are in different components. The family is finite, hence has an inclusion-minimal member S .

If $|S| \leq 1$, then S is a clique and we are done. So assume $|S| \geq 2$ and let x, y be two distinct vertices of S . Let A be the vertex set of the component of $G - S$ containing a , and B that of the component containing b ; by the choice of S we have $A \neq B$. There is no edge of G between A and B : such an edge would join two vertices of $V(G) \setminus S$, hence would be an edge of $G - S$ between two different components, which is impossible.

Step 1. (Every vertex of S has a neighbour in A and a neighbour in B .) Let $s \in S$.

By minimality of S , the set $S \setminus \{s\}$ is not in the family, so $G - (S \setminus \{s\})$ contains a path P from a to b . Since $G - S$ has no path from a to b , the path P contains s ; as P is a path, s occurs on it exactly once. Let s' be the vertex just before s on P and s'' the vertex just after. The portion of P from a to s' avoids S entirely, because P avoids $S \setminus \{s\}$ and does not reach s before s' ; hence s' lies in the same component of $G - S$ as a , that is $s' \in A$, and $s's \in E(G)$. Symmetrically $s'' \in B$ and $s''s \in E(G)$.

Step 2. (A path from x to y through A , and one through B .) By Step 1, x has a neighbour $x_A \in A$ and y has a neighbour $y_A \in A$. Since $G[A]$ is connected (it is the graph induced by a component of $G - S$), it contains a path from x_A to y_A . Prefixing x and suffixing y gives a walk from x to y whose internal vertices all lie in A ; by Lemma 2.1 it contains a path from x to y whose internal vertices all lie in A . Among all such paths, let P_A be one of minimum length. Since x and y are non-adjacent, P_A has length at least 2. Symmetrically, let P_B be a minimum-length path from x to y whose internal vertices all lie in B ; it also has length at least 2.

Step 3. (P_A and P_B are induced and meet only in x, y .) The internal vertices of P_A lie in A , those of P_B lie in B , and $A \cap B = \emptyset$; also $x, y \in S$, so $x, y \notin A \cup B$. Hence P_A and P_B share exactly the vertices x and y . Moreover P_A is an induced path: if two of its vertices were adjacent but non-consecutive on P_A , then deleting the vertices strictly between them along P_A (as in Lemma 4.2) would give a shorter walk from x to y , hence a shorter path from x to y with all internal vertices still in A , contradicting the choice of P_A . The same holds for P_B .

Step 4. (A chordless cycle.) Let Z be the cycle that follows P_A from x to y and then P_B back from y to x . By Step 3 its vertices are distinct, and its length is $|P_A| + |P_B| \geq 4$. We show Z has no chord. Let p, q be two vertices of Z that are non-consecutive on Z .

- If p, q both lie on P_A : they are non-consecutive on P_A as well, so $pq \notin E(G)$ because P_A is induced (Step 3).
- If p, q both lie on P_B : likewise $pq \notin E(G)$.
- If one of p, q is an internal vertex of P_A (hence in A) and the other is an internal vertex of P_B (hence in B): $pq \notin E(G)$ because there is no edge of G between A and B .
- If $\{p, q\} = \{x, y\}$: $pq \notin E(G)$ by hypothesis.

Every pair of non-consecutive vertices of Z falls into one of these cases (each of p, q lies on P_A or on P_B , and the vertices x, y lie on both). So Z is a chordless cycle of length at least 4 in G , contradicting that G is chordal.

The contradiction shows that the assumption “ x, y non-adjacent” is untenable. Hence every two vertices of S are adjacent, that is S is a clique. $\square$

Lemma 4.4 (Existence of a simplicial vertex). *Every chordal graph with at least one vertex has a simplicial vertex. Moreover, every chordal graph that is not a complete graph has two non-adjacent simplicial vertices.*

Proof. Induction on $n = |V(G)|$.

Base $n = 1$. The unique vertex has empty neighbourhood, a clique, so it is simplicial. A one-vertex graph is complete, so the second statement is vacuous.

Inductive step $n \geq 2$. Assume both statements for all chordal graphs with fewer than n vertices.

Case 1: G is complete. Every vertex v has $N_G(v) = V(G) \setminus \{v\}$, a clique, so v is simplicial. The first statement holds; the second is vacuous.

Case 2: G is not complete. Fix distinct non-adjacent vertices a, b . By Lemma 4.3 there is an inclusion-minimal $S \subseteq V(G) \setminus \{a, b\}$ separating a from b , and S is a clique. Let A be the vertex set of the component of $G - S$ containing a , and B that of the component containing b .

Claim: G has a simplicial vertex lying in A . Let $H = G[A \cup S]$. By Lemma 4.1, H is chordal. Since $b \notin A \cup S$ we have $|V(H)| \leq n - 1$, and $A \ni a$ gives $|V(H)| \geq 1$; so $1 \leq |V(H)| < n$.

- If H is complete, then $A \cup S$ is a clique. Pick any $v \in A$. Every neighbour of v in G lies in $A \cup S$: a neighbour of v is either in the same component of $G - S$ as v , namely in A , or in S . Hence $N_G(v) \subseteq (A \cup S) \setminus \{v\}$, a subset of the clique $A \cup S$, so $N_G(v)$ is a clique and v is simplicial in G .
- If H is not complete, then by the inductive hypothesis H has two non-adjacent simplicial vertices u_1, u_2 . Since S is a clique and u_1, u_2 are non-adjacent, at most one of them lies in S ; so at least one, say u_1 , lies in A . As above, $N_G(u_1) \subseteq A \cup S = V(H)$, and H is an induced subgraph of G , so $N_G(u_1) = N_H(u_1)$, which is a clique because u_1 is simplicial in H . Hence u_1 is simplicial in G and lies in A .

This proves the claim; call the resulting vertex $s_A \in A$.

By the same argument applied to $G[B \cup S]$, there is a simplicial vertex s_B of G lying in B . Since $s_A \in A$ and $s_B \in B$ lie in different components of $G - S$, they are non-adjacent in G (an edge $s_A s_B$ would be an edge of G between A and B). This proves the second statement, and the first follows because s_A is a simplicial vertex of G . $\square$

5 A marking process and its soundness

We now describe a propagation process on the vertices of a graph. Section 6 proves it decides every chordal graph; section 7 adapts it to digraphs.

Definition 5.1 (The marking process for a graph). Let G be a graph. A *state* is a function $\sigma: V(G) \rightarrow \{\text{IN}, \text{OUT}, \text{UNKNOWN}\}$. The *initial state* σ_0 assigns **UNKNOWN** to every vertex. Given a state σ , a single *step* produces a new state by applying one of the following rules to a vertex meeting its precondition.

- (Spread)** Choose v with $\sigma(v) = \text{IN}$ and a vertex $u \in N_G(v)$ with $\sigma(u) = \text{UNKNOWN}$. The new state equals σ except that u is reassigned to **OUT**.
- (Force)** Choose v with $\sigma(v) = \text{UNKNOWN}$ such that the set

$$C_\sigma(v) := \{u \in N_G(v) : \sigma(u) \neq \text{OUT}\}$$

is a clique of G (recall a set of size at most 1 is a clique). The new state equals σ except that v is reassigned to **IN**.

A *run* is a sequence $\sigma_0, \sigma_1, \sigma_2, \dots$ in which each σ_{t+1} is obtained from σ_t by one step. A state is *terminal* if neither rule can be applied to it. A run is *complete* if it is finite and its last state is terminal.

Lemma 5.2 (Termination). *Every run has length at most $n = |V(G)|$, and every run can be extended to a complete run.*

Proof. Each step strictly decreases the number of vertices in state UNKNOWN: (Spread) changes one vertex from UNKNOWN to OUT, and (Force) changes one vertex from UNKNOWN to IN; no rule ever produces a UNKNOWN. There are n vertices, so at most n steps occur. If a run has not reached a terminal state, some rule applies and the run can be extended by one step; since this cannot continue beyond n steps, the run reaches a terminal state. $\square$

Lemma 5.3 (Soundness). *Let $\sigma_0, \sigma_1, \dots, \sigma_T$ be a run for a graph G , and let S be any 2-kernel of G . Then for every vertex v and every t : if $\sigma_t(v) = \text{IN}$ then $v \in S$, and if $\sigma_t(v) = \text{OUT}$ then $v \notin S$.*

Proof. Once a vertex leaves state UNKNOWN its value never changes again, because no rule alters a vertex already IN or OUT. So it suffices to show that at the step where a vertex v is reassigned away from UNKNOWN, its new value is correct for S . We induct on the step index and may assume the statement for every vertex reassigned at an earlier step.

Reassignment by (Spread). Here v becomes OUT because some $u \in N_G(v)$ has $\sigma(u) = \text{IN}$ at that moment. The vertex u was reassigned to IN at an earlier step, so by the inductive hypothesis $u \in S$. Since $u \in N_G(v)$, the edge uv has both ends in $\{u, v\}$; as $u \in S$ and S is independent, $v \notin S$.

Reassignment by (Force). Here v becomes IN because $C := C_\sigma(v)$ is a clique of G . Suppose for contradiction that $v \notin S$. Let $w \in N_G(v) \cap S$. If $\sigma(w) = \text{OUT}$ at this moment, then w was reassigned to OUT earlier, so by the inductive hypothesis $w \notin S$, a contradiction; hence $\sigma(w) \neq \text{OUT}$, so $w \in C$. Thus $N_G(v) \cap S \subseteq C$. Because $v \notin S$ and S is a 2-kernel of the graph G , we have $|N_G(v) \cap S| \geq 2$; pick distinct $w_1, w_2 \in N_G(v) \cap S \subseteq C$. Since $w_1, w_2 \in S$ and S is independent, $w_1 w_2 \notin E(G)$; but $w_1, w_2 \in C$ and C is a clique, so $w_1 w_2 \in E(G)$. This contradiction shows $v \in S$. $\square$

6 2-Kernels of chordal graphs

Theorem 6.1. *Let G be a chordal graph and let $\sigma_0, \dots, \sigma_T$ be a complete run of the marking process for G (Definition 5.1). Then:*

- (1) σ_T assigns UNKNOWN to no vertex.
- (2) For every 2-kernel S of G and every vertex v : $v \in S$ if and only if $\sigma_T(v) = \text{IN}$. Consequently G has at most one 2-kernel; writing $S^* = \{v : \sigma_T(v) = \text{IN}\}$, the graph G has a 2-kernel if and only if S^* is a 2-kernel, and then S^* is the unique one.
- (3) Whether G has a 2-kernel, and the 2-kernel itself when it exists, can be determined in time polynomial in $n = |V(G)|$.

Proof. (1) Suppose $\sigma_T(w) = \text{UNKNOWN}$ for some w , and let $W = \{v : \sigma_T(v) = \text{UNKNOWN}\}$, so $w \in W$ and $W \neq \emptyset$.

Step 1. No vertex of W has a neighbour u with $\sigma_T(u) = \text{IN}$: otherwise (Spread) could be applied to the pair (that neighbour u , the vertex of W), contradicting that σ_T is terminal.

Step 2. Fix $v \in W$. By Step 1 every neighbour of v is OUT or UNKNOWN, so

$$C_{\sigma_T}(v) = \{u \in N_G(v) : \sigma_T(u) \neq \text{OUT}\} = N_G(v) \cap W.$$

Step 3. By Lemma 4.1 the graph $G[W]$ is chordal, and $W \neq \emptyset$, so by Lemma 4.4 it has a simplicial vertex x . Then $N_{G[W]}(x) = N_G(x) \cap W$ is a clique of $G[W]$; since $G[W]$ is an induced subgraph of G , it is a clique of G .

Step 4. By Step 2 applied to x , we have $C_{\sigma_T}(x) = N_G(x) \cap W$, which is a clique of G by Step 3. Hence (Force) applies to x (which is in state UNKNOWN), contradicting that σ_T is terminal.

The contradiction shows $W = \emptyset$, proving (1).

(2) Let S be a 2-kernel of G . By (1), every vertex is IN or OUT in σ_T . By Lemma 5.3, $\sigma_T(v) = \text{IN}$ implies $v \in S$ and $\sigma_T(v) = \text{OUT}$ implies $v \notin S$. Since every vertex is IN or OUT, the converse implications also hold: if $v \in S$ then $\sigma_T(v) \neq \text{OUT}$, so $\sigma_T(v) = \text{IN}$; if $v \notin S$ then $\sigma_T(v) \neq \text{IN}$, so $\sigma_T(v) = \text{OUT}$. Hence $S = \{v : \sigma_T(v) = \text{IN}\} = S^*$. So any 2-kernel equals S^* : there is at most one, and one exists precisely when S^* itself is a 2-kernel.

(3) By Lemma 5.2 a complete run has at most n steps. Deciding which rule applies to which vertex, and testing whether a set $C_\sigma(v)$ is a clique, are computations polynomial in n ; so a complete run is produced in polynomial time. Then test directly from the definition whether S^* is independent and 2-dominating, again in polynomial time. By (2), G has a 2-kernel if and only if this test succeeds, and then S^* is the unique 2-kernel. $\square$

7 2-Kernels of digraphs with a chordal underlying graph

We adapt Definition 5.1 to a digraph D . The only change is that (Force) inspects the *out*-neighbourhood, while (Spread) still uses the full underlying neighbourhood, since independence forbids an arc between two chosen vertices in either direction.

Definition 7.1 (The marking process for a digraph). Let D be a digraph with underlying graph $U(D)$. States and the initial state are as in Definition 5.1, on the vertex set $V(D)$. A single step applies one of:

(Spread') Choose v with $\sigma(v) = \text{IN}$ and $u \in N_D(v)$ with $\sigma(u) = \text{UNKNOWN}$; reassign u to OUT.

(Force') Choose v with $\sigma(v) = \text{UNKNOWN}$ such that

$$C'_\sigma(v) := \{u \in N_D^+(v) : \sigma(u) \neq \text{OUT}\}$$

is a clique of $U(D)$; reassign v to IN.

Runs, terminal states, and complete runs are defined as before.

Lemma 7.2 (Termination, digraph version). *Every run of the process of Definition 7.1 has length at most $|V(D)|$, and every run extends to a complete run.*

Proof. Identical to Lemma 5.2: each step turns one vertex from UNKNOWN into OUT or IN, and no step creates a UNKNOWN. $\square$

Lemma 7.3 (Soundness, digraph version). *Let $\sigma_0, \dots, \sigma_T$ be a run of the process of Definition 7.1 for a digraph D , and let S be any 2-kernel of D . Then for every vertex v and every t : if $\sigma_t(v) = \text{IN}$ then $v \in S$, and if $\sigma_t(v) = \text{OUT}$ then $v \notin S$.*

Proof. As in Lemma 5.3, a vertex leaving UNKNOWN never changes again, and we induct on the step index.

Reassignment by (Spread'). Here v becomes OUT because some $u \in N_D(v)$ has $\sigma(u) = \text{IN}$, with u reassigned earlier, so $u \in S$ by the inductive hypothesis. Since $u \in N_D(v)$, at least one of $(u, v), (v, u)$ is an arc of D ; as $u \in S$ and S is independent, $v \notin S$.

Reassignment by (Force'). Here v becomes **IN** because $C := \{u \in N_D^+(v) : \sigma(u) \neq \text{OUT}\}$ is a clique of $U(D)$. Suppose $v \notin S$. Let $w \in N_D^+(v) \cap S$. If $\sigma(w) = \text{OUT}$, then w was reassigned earlier, so $w \notin S$ by the inductive hypothesis, a contradiction; hence $\sigma(w) \neq \text{OUT}$ and $w \in C$. Thus $N_D^+(v) \cap S \subseteq C$. Because $v \notin S$ and S is a 2-kernel of D , we have $|N_D^+(v) \cap S| \geq 2$; pick distinct $w_1, w_2 \in N_D^+(v) \cap S \subseteq C$. Since $w_1, w_2 \in S$ and S is independent, $w_1 w_2 \notin E(U(D))$; but $w_1, w_2 \in C$ and C is a clique of $U(D)$, so $w_1 w_2 \in E(U(D))$. This contradiction shows $v \in S$. $\square$

Theorem 7.4. *Let D be a digraph whose underlying graph $U(D)$ is chordal, and let $\sigma_0, \dots, \sigma_T$ be a complete run of the process of Definition 7.1. Then:*

- (1) σ_T assigns **UNKNOWN** to no vertex.
- (2) For every 2-kernel S of D and every vertex v : $v \in S$ if and only if $\sigma_T(v) = \text{IN}$. Consequently D has at most one 2-kernel, and, with $S^* = \{v : \sigma_T(v) = \text{IN}\}$, the digraph D has a 2-kernel if and only if S^* is one, and then it is the unique one.
- (3) Whether D has a 2-kernel, and the 2-kernel itself when it exists, can be determined in time polynomial in $|V(D)|$. No hypothesis on out-degrees is used anywhere in the argument.

Proof. Key observation. For any vertex x and any $W \subseteq V(D)$: if $N_D(x) \cap W$ is a clique of $U(D)$, then $N_D^+(x) \cap W$ is also a clique of $U(D)$. Indeed, $N_D^+(x) \subseteq N_D(x)$ by (2.1), so $N_D^+(x) \cap W \subseteq N_D(x) \cap W$, and any subset of a clique is a clique.

(1) Suppose $W := \{v : \sigma_T(v) = \text{UNKNOWN}\} \neq \emptyset$.

Step 1. No vertex of W has an underlying neighbour u with $\sigma_T(u) = \text{IN}$: otherwise (Spread') would apply, contradicting that σ_T is terminal.

Step 2. Fix $v \in W$. By Step 1 every underlying neighbour of v is **OUT** or in W . Since $N_D^+(v) \subseteq N_D(v)$, it follows that $C'_{\sigma_T}(v) = \{u \in N_D^+(v) : \sigma_T(u) \neq \text{OUT}\} = N_D^+(v) \cap W$.

Step 3. By Lemma 4.1 the graph $U(D)[W]$ is chordal, and $W \neq \emptyset$, so by Lemma 4.4 it has a simplicial vertex x : the set $N_D(x) \cap W$, which is the neighbourhood of x in $U(D)[W]$, is a clique of $U(D)[W]$, hence of $U(D)$.

Step 4. By the key observation, $N_D^+(x) \cap W$ is a clique of $U(D)$. By Step 2 this set equals $C'_{\sigma_T}(x)$. Hence (Force') applies to x , contradicting that σ_T is terminal.

So $W = \emptyset$, proving (1).

(2) Identical to the proof of Theorem 6.1(2), with Lemma 7.3 in place of Lemma 5.3.

(3) Identical to the proof of Theorem 6.1(3): a complete run has at most $|V(D)|$ steps by Lemma 7.2, each step and each clique test is polynomial, and S^* is checked against the definition of a 2-kernel of D . The argument uses N_D^+ , N_D , and chordality of $U(D)$ only; it never refers to d_D^+ or $\delta^+(D)$. $\square$

Remark 7.5. If G is a graph and D is its associated symmetric digraph, then $N_D^+(v) = N_D(v) = N_G(v)$ and $U(D) = G$, so Definition 7.1 becomes exactly Definition 5.1 and Lemma 7.3 becomes Lemma 5.3. Thus Theorem 6.1 is the special case of Theorem 7.4 in which D is symmetric. Theorem 7.4 is strictly more general, since it also applies to digraphs that are not symmetric, for example to orientations of chordal graphs.

8 A lower bound for oriented graphs

Theorem 8.1. *Let D be an oriented graph with $\delta^+(D) \geq 2$. If D has a 2-kernel, then $|V(D)| \geq 8$.*

Proof. Let J be a 2-kernel of D . Write $n = |V(D)|$, $j = |J|$, $B = V(D) \setminus J$, and $t = |B| = n - j$.

Step 1. ($t \geq 1$ and $j \geq 1$.) If $t = 0$ then $J = V(D)$ is independent, so D has no arcs; but $\delta^+(D) \geq 2$ and $V(D) \neq \emptyset$ force some vertex to have out-degree at least 2, a contradiction. So $t \geq 1$. If $j = 0$ then $J = \emptyset$, and every vertex of $V(D)$ lies outside J and would need $|N_D^+(v) \cap \emptyset| \geq 2$, that is $0 \geq 2$, impossible since $V(D) \neq \emptyset$. So $j \geq 1$.

Step 2. (Lower bound on the number of arcs from B to J .) For $x \in J$ let $d_x = |\{b \in B : (b, x) \in A(D)\}|$. The number of arcs from B into J equals $\sum_{x \in J} d_x$. On the other hand, each $b \in B$ lies outside J , so 2-domination gives $|N_D^+(b) \cap J| \geq 2$; summing over $b \in B$, the number of arcs from B into J is $\sum_{b \in B} |N_D^+(b) \cap J| \geq 2t$. Hence

$$\sum_{x \in J} d_x \geq 2t. \quad (8.1)$$

Step 3. (Upper bound on each d_x .) Fix $x \in J$. Since J is independent, x has no arc to another vertex of J , so $N_D^+(x) \subseteq B$. Let $P_x = \{b \in B : (b, x) \in A(D)\}$, so $|P_x| = d_x$. For $b \in P_x$ the pair (b, x) is an arc, and since D is oriented, (x, b) is not an arc; hence $b \notin N_D^+(x)$. Therefore $N_D^+(x) \subseteq B \setminus P_x$, so

$$d_D^+(x) \leq |B \setminus P_x| = t - d_x.$$

Since $d_D^+(x) \geq \delta^+(D) \geq 2$, we get $2 \leq t - d_x$, that is

$$d_x \leq t - 2 \quad \text{for every } x \in J. \quad (8.2)$$

Step 4. (Combining.) From (8.1) and (8.2),

$$2t \leq \sum_{x \in J} d_x \leq \sum_{x \in J} (t - 2) = j(t - 2). \quad (8.3)$$

The left side of (8.3) is positive and $j \geq 1$, so $t - 2 > 0$, that is $t \geq 3$. Dividing (8.3) by $t - 2 > 0$ gives

$$j \geq \frac{2t}{t - 2}. \quad (8.4)$$

Step 5. (Minimising $n = j + t$.) Here j and t are integers with $t \geq 3$ and $j \geq \lceil 2t/(t - 2) \rceil$ by (8.4). We bound $n = j + t$ from below for each t :

$$\begin{aligned} t = 3 : & \quad 2t/(t - 2) = 6, & j \geq 6, \quad n \geq 9; \\ t = 4 : & \quad 2t/(t - 2) = 4, & j \geq 4, \quad n \geq 8; \\ t = 5 : & \quad 2t/(t - 2) = 10/3, & j \geq 4, \quad n \geq 9; \\ t = 6 : & \quad 2t/(t - 2) = 3, & j \geq 3, \quad n \geq 9; \\ t \geq 7 : & \quad 2t/(t - 2) = 2 + \frac{4}{t - 2} > 2, \quad j \geq 3, \quad n = j + t \geq 3 + 7 = 10. \end{aligned}$$

In every case $n \geq 8$. Hence $|V(D)| \geq 8$. □

Example 8.2. The bound in Theorem 8.1 is attained. Let D have vertex set $\{0, 1, 2, 3, 4, 5, 6, 7\}$ and the out-neighbourhoods

$$\begin{aligned} N^+(0) &= \{5, 6\}, & N^+(1) &= \{6, 7\}, & N^+(2) &= \{4, 7\}, & N^+(3) &= \{4, 5\}, \\ N^+(4) &= \{0, 1\}, & N^+(5) &= \{1, 2\}, & N^+(6) &= \{2, 3\}, & N^+(7) &= \{0, 3\}. \end{aligned}$$

Then:

- (i) Every vertex has out-degree exactly 2, so $\delta^+(D) = 2$.
- (ii) D is an oriented graph. For each listed arc (u, v) one checks that $u \notin N^+(v)$, so (v, u) is not an arc. For instance $(0, 5)$ is an arc and $N^+(5) = \{1, 2\}$ does not contain 0; the other fifteen checks are identical.
- (iii) $J = \{0, 1, 2, 3\}$ is a 2-kernel. It is independent because $N^+(0), N^+(1), N^+(2), N^+(3)$ are all contained in $\{4, 5, 6, 7\}$, so no arc has both ends in J . It is 2-dominating because the vertices outside J are 4, 5, 6, 7 and

$$N^+(4) = \{0, 1\} \subseteq J, \quad N^+(5) = \{1, 2\} \subseteq J, \quad N^+(6) = \{2, 3\} \subseteq J, \quad N^+(7) = \{0, 3\} \subseteq J,$$

each of size 2.

Thus D is an oriented graph on 8 vertices with $\delta^+(D) = 2$ that has a 2-kernel, so 8 is the least possible number of vertices. One checks in the same way that $\{4, 5, 6, 7\}$ is also a 2-kernel of D .

9 2-Kernels of acyclic digraphs

Lemma 9.1 (Sinks exist). *Every non-empty acyclic digraph D has a vertex z with $N_D^+(z) = \emptyset$.*

Proof. Suppose every vertex has out-degree at least 1. Let $n = |V(D)| \geq 1$. Choose any $u_0 \in V(D)$ and, for $i \geq 0$, choose $u_{i+1} \in N_D^+(u_i)$. Among $u_0, u_1, \dots, u_n$ there are $n + 1$ terms from a set of size n , so $u_a = u_b$ for some $0 \leq a < b \leq n$. Then $u_a, u_{a+1}, \dots, u_b$ is a closed directed walk of length $b - a \geq 1$. Among all pairs (p, q) with $a \leq p < q \leq b$ and $u_p = u_q$, choose one with $q - p$ minimum. Then $u_p, u_{p+1}, \dots, u_{q-1}$ are pairwise distinct, so $u_p \rightarrow u_{p+1} \rightarrow \dots \rightarrow u_{q-1} \rightarrow u_q = u_p$ is a directed cycle of D , contradicting acyclicity. Hence some vertex has out-degree 0. $\square$

Lemma 9.2 (Topological order). *For every acyclic digraph D with $|V(D)| = n$ there is an ordering $v_1, v_2, \dots, v_n$ of $V(D)$ such that every arc (v_i, v_j) satisfies $i < j$.*

Proof. Induction on n . For $n = 0$ the empty ordering works. For $n \geq 1$, by Lemma 9.1 choose z with $N_D^+(z) = \emptyset$. The digraph $D - z$ obtained by deleting z and its incident arcs is acyclic, since any directed cycle of $D - z$ is one of D . By the inductive hypothesis $D - z$ has an ordering $w_1, \dots, w_{n-1}$ with every arc increasing. Set $v_i = w_i$ for $1 \leq i \leq n - 1$ and $v_n = z$. Arcs among $w_1, \dots, w_{n-1}$ are increasing by hypothesis; an arc $(v_i, v_n) = (v_i, z)$ has $i < n$; and there is no arc $(v_n, v_j) = (z, v_j)$ because $N_D^+(z) = \emptyset$. So every arc is increasing. $\square$

Definition 9.3 (Backward sweep). Let D be an acyclic digraph with a fixed topological order $v_1, \dots, v_n$ (Lemma 9.2). Process the indices $i = n, n - 1, \dots, 1$ in this order. When index i is processed, every out-neighbour v_j of v_i has $j > i$, so $\sigma(v_j)$ is already defined; set

$$k_i = |\{j : (v_i, v_j) \in A(D) \text{ and } \sigma(v_j) = \text{IN}\}|.$$

Then:

- if $k_i = 0$, set $\sigma(v_i) = \text{IN}$;
- if $k_i = 1$, stop and declare *failure*;
- if $k_i \geq 2$, set $\sigma(v_i) = \text{OUT}$.

If index 1 is processed without failure, the sweep *succeeds*; put $S^* = \{v_i : \sigma(v_i) = \text{IN}\}$.

Theorem 9.4. *Let D be an acyclic digraph with a fixed topological order, and run the backward sweep of Definition 9.3.*

- (1) If D has a 2-kernel S , then the sweep succeeds and $S^* = S$.
- (2) If the sweep succeeds, then S^* is a 2-kernel of D .
- (3) Hence D has a 2-kernel if and only if the sweep succeeds, and in that case S^* is the unique 2-kernel of D . In particular D has at most one 2-kernel.

Proof. (1) Let S be a 2-kernel of D . For $i \in \{n, n-1, \dots, 1\}$ let $P(i)$ be the statement:

the sweep does not fail while processing any index in $\{i, i+1, \dots, n\}$, and for every $j \in \{i, \dots, n\}$ we have $\sigma(v_j) = \text{IN}$ if and only if $v_j \in S$.

We prove $P(i)$ by downward induction on i . The hypothesis $P(i+1)$ is empty when $i = n$. Assume $P(i+1)$ and process index i .

Every out-neighbour v_j of v_i has $j > i$, hence $j \in \{i+1, \dots, n\}$; so by $P(i+1)$, $\sigma(v_j) = \text{IN}$ if and only if $v_j \in S$. Therefore

$$k_i = |\{j : (v_i, v_j) \in A(D), \sigma(v_j) = \text{IN}\}| = |\{j : (v_i, v_j) \in A(D), v_j \in S\}| = |N_D^+(v_i) \cap S|.$$

By Lemma 2.2(c), $|N_D^+(v_i) \cap S| \neq 1$, so $k_i \neq 1$ and the sweep does not fail at i .

- If $k_i = 0$: the sweep sets $\sigma(v_i) = \text{IN}$; and $|N_D^+(v_i) \cap S| = 0$, so $v_i \in S$ by Lemma 2.2(d). Both sides of the equivalence hold.
- If $k_i \geq 2$: the sweep sets $\sigma(v_i) = \text{OUT}$; and $|N_D^+(v_i) \cap S| = k_i \geq 1$, so $v_i \notin S$ by Lemma 2.2(a). Both sides of the equivalence fail.

Together with $P(i+1)$ this gives $P(i)$. Hence $P(1)$ holds: the sweep never fails, so it succeeds, and $\sigma(v_j) = \text{IN}$ if and only if $v_j \in S$ for all j , that is $S^* = S$.

(2) Suppose the sweep succeeds and $S^* = \{v_i : \sigma(v_i) = \text{IN}\}$.

Independence. Suppose some arc (v_i, v_j) has both ends in S^* . Since it is an arc and the order is topological, $i < j$, so v_j was processed before v_i , and $\sigma(v_j) = \text{IN}$ (as $v_j \in S^*$) held when v_i was processed. Hence $k_i \geq 1$. But $\sigma(v_i) = \text{IN}$ (as $v_i \in S^*$) means the sweep took the branch $k_i = 0$ at index i , so $k_i = 0$: a contradiction. So no arc has both ends in S^* .

2-domination. Let $v_i \notin S^*$, so $\sigma(v_i) = \text{OUT}$, which occurs only in the branch $k_i \geq 2$. When index i was processed, at least two out-neighbours v_j of v_i had $\sigma(v_j) = \text{IN}$; each such v_j lies in S^* , and its status is never changed afterwards. Hence $|N_D^+(v_i) \cap S^*| \geq 2$.

So S^* is independent and 2-dominating, that is a 2-kernel of D .

(3) If D has a 2-kernel S , then by (1) the sweep succeeds and $S = S^*$; so a 2-kernel, if any exists, equals S^* and is unique. If the sweep succeeds, then by (2) S^* is a 2-kernel. Hence D has a 2-kernel exactly when the sweep succeeds, and then S^* is the only one. $\square$

Corollary 9.5. *Whether an acyclic digraph D has a 2-kernel, and the 2-kernel itself when it exists, can be computed in time $O(n + m)$, where $n = |V(D)|$ and $m = |A(D)|$.*

Proof. A topological order is obtained in $O(n + m)$ time by repeatedly removing a vertex of out-degree 0 (Lemma 9.1), maintaining current out-degrees as arcs are deleted and keeping the out-degree-0 vertices in a queue. The backward sweep of Definition 9.3 processes each vertex once and, for vertex v_i , inspects each arc leaving v_i once in order to compute k_i ; the total work is $O(n + m)$. By Theorem 9.4(3) the outcome of the sweep is itself the answer, so no separate verification step is needed. $\square$

Remark 9.6. Although the backward sweep is defined relative to a chosen topological order, its outcome does not depend on the choice. If D has a 2-kernel, then for every topological order the sweep succeeds with S^* equal to that unique 2-kernel, by Theorem 9.4(1). If D has no 2-kernel, then for every topological order the sweep fails, since a successful sweep would produce a 2-kernel by Theorem 9.4(2).

10 2-Kernels of subdivisions

Definition 10.1 (Subdivision). Let G be a graph. Its *subdivision* $S(G)$ is the graph with vertex set

$$V(S(G)) = V(G) \cup \{x_e : e \in E(G)\},$$

where the x_e are new, pairwise distinct vertices, one for each edge, and edge set

$$E(S(G)) = \{ux_e, x_e w : e = uw \in E(G)\}.$$

Thus each edge uw of G is replaced by the path u, x_{uw}, w of length 2.

Theorem 10.2. *For every graph G , the set $V(G)$ is a 2-kernel of $S(G)$.*

Proof. We treat $S(G)$ as a graph, so that 2-domination of a vertex v means $|N_{S(G)}(v) \cap V(G)| \geq 2$.

Step 1. (Independence.) Let $u, w \in V(G)$. By Definition 10.1, the neighbours of u in $S(G)$ are exactly the vertices x_e for edges e of G incident with u ; every such x_e is a new vertex and does not lie in $V(G)$. Hence u and w are non-adjacent in $S(G)$. So $V(G)$ is independent in $S(G)$.

Step 2. (2-domination.) Let $v \in V(S(G)) \setminus V(G)$. Then $v = x_e$ for some edge $e = uw$ of G , and by Definition 10.1, $N_{S(G)}(x_e) = \{u, w\} \subseteq V(G)$ with $u \neq w$. Hence $|N_{S(G)}(v) \cap V(G)| = 2$.

By Steps 1 and 2, $V(G)$ is independent and 2-dominating in $S(G)$, hence a 2-kernel. If $E(G) = \emptyset$ then $S(G) = G$, there is no vertex outside $V(G)$, and $V(G)$ is a 2-kernel of $S(G)$ with the 2-domination condition holding vacuously. $\square$